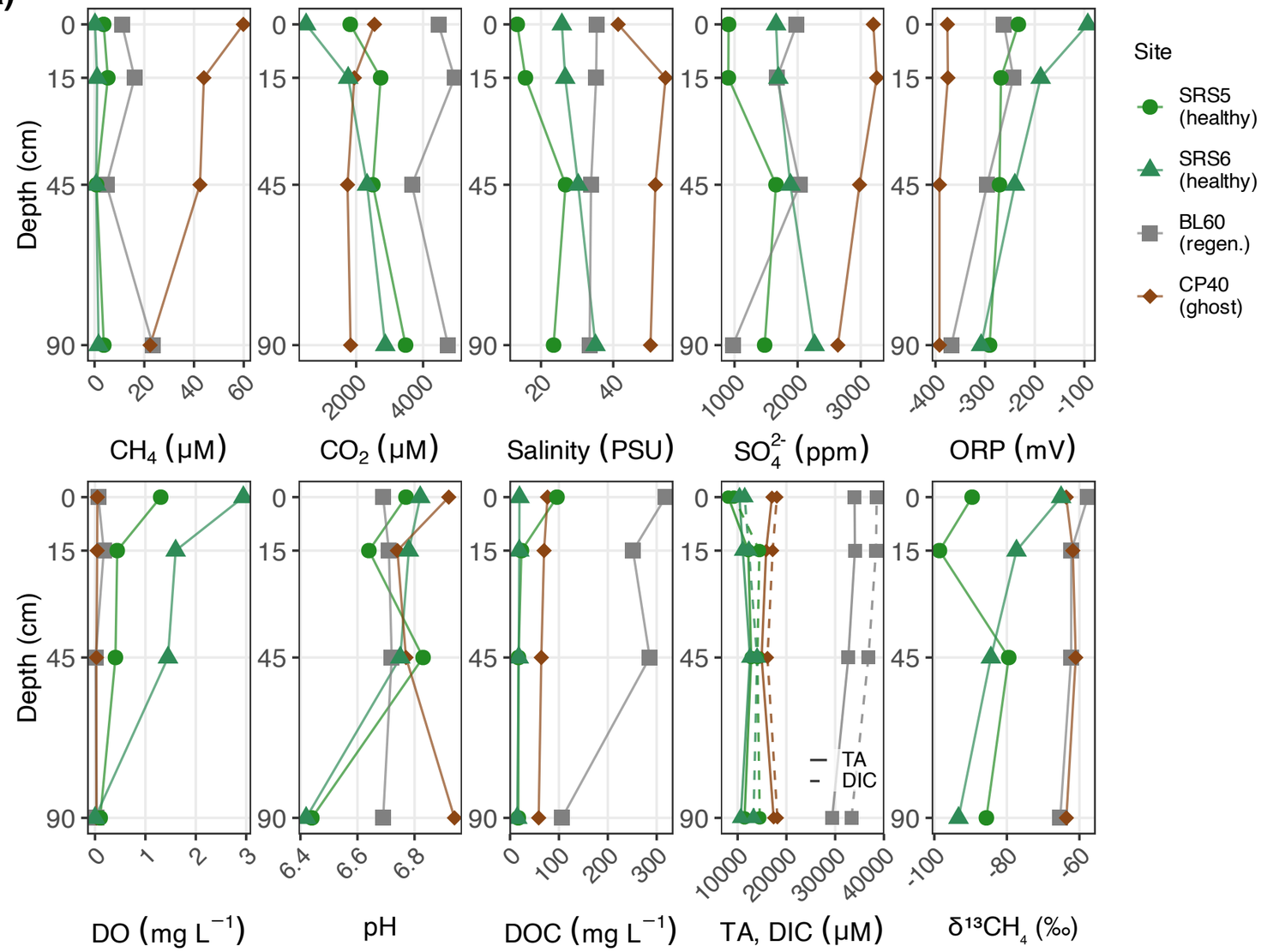
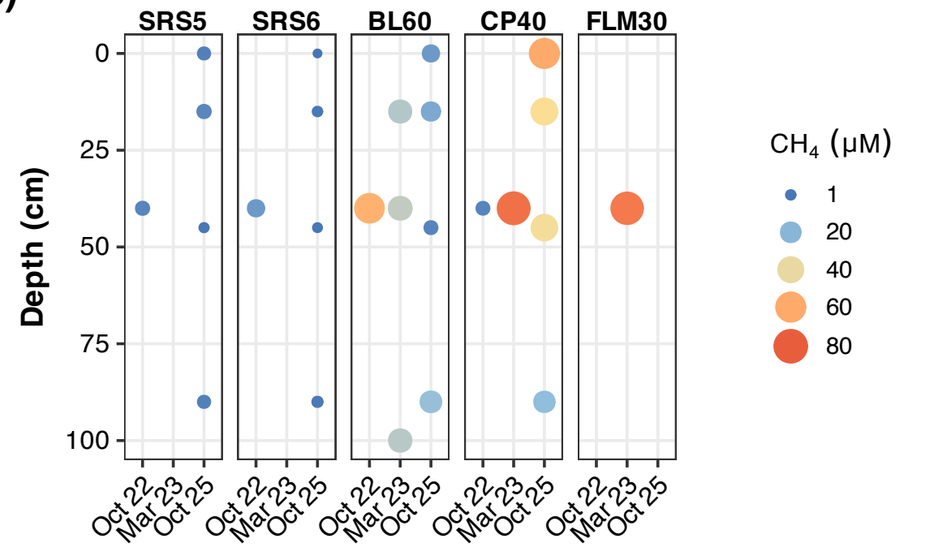
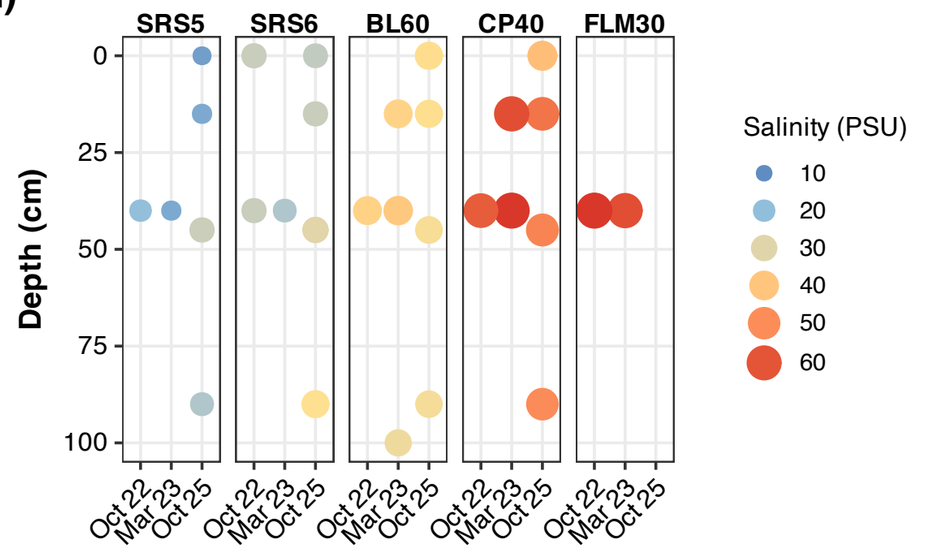
(a)


Depth (cm)
CH4 (μM)
CO2 (μM)
Salinity (PSU)
SO4 2- (ppm)
ORP (mV)
DO (mg L−1)
pH
DOC (mg L−1)
TA, DIC (μM)
δ13CH4 (‰)
TA
DIC
Site
SRS5 (healthy)
SRS6 (healthy)
BL60 (regen.)
CP40 (ghost)

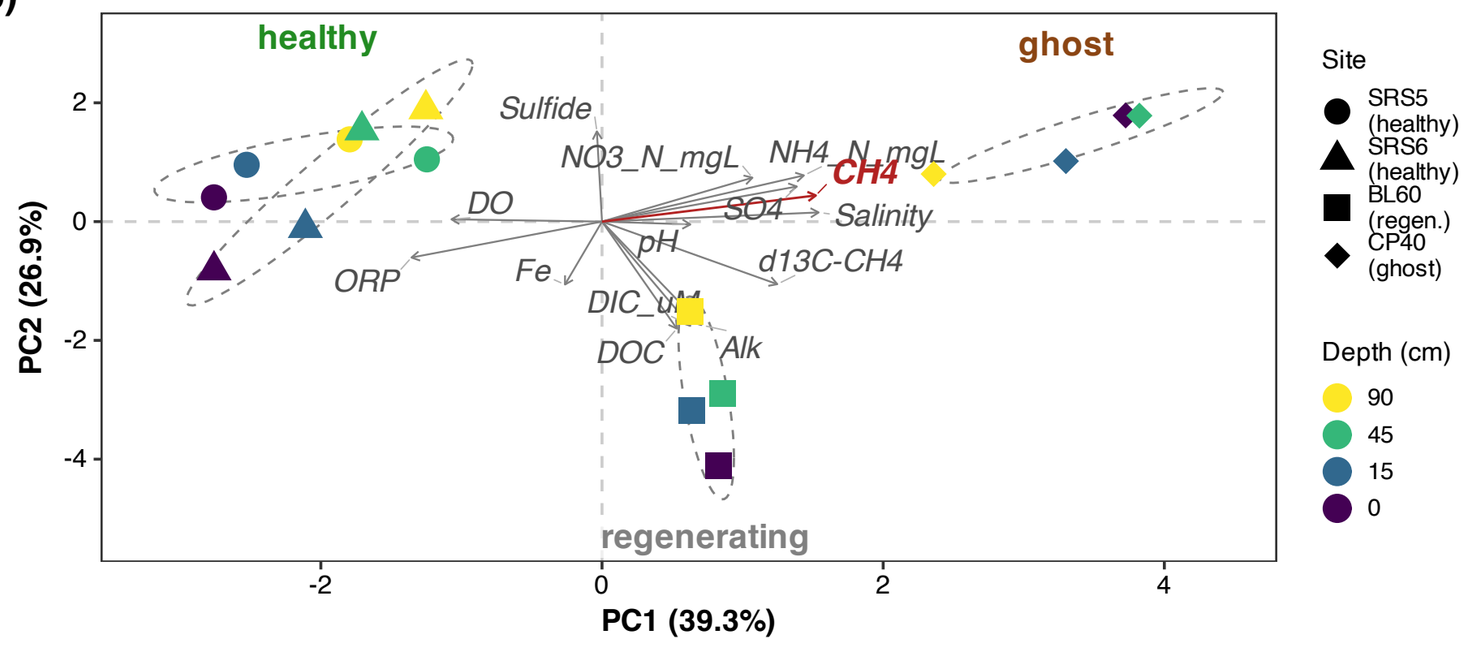
(c)


SRS5
SRS6
BL60
CP40
FLM30
Depth (cm)
Oct 22
Mar 23
Oct 25
CH4 (μM)
1
20
40
60
80

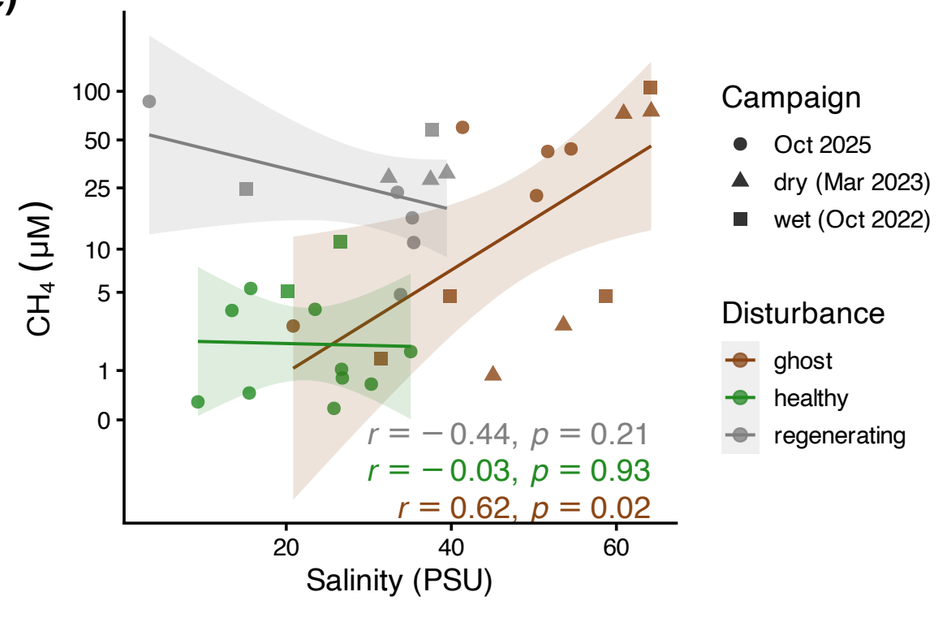
(d)


SRS5
SRS6
BL60
CP40
FLM30
Depth (cm)
Oct 22
Mar 23
Oct 25
Salinity (PSU)
10
20
30
40
50
60

(b)


healthy
ghost
regenerating
Sulfide
NO3_N_mgL
NH4_N_mgL
CH4
DO
SO4
Salinity
pH
d13C-CH4
ORP
Fe
DIC_uM
DOC
Alk
PC1 (39.3%)
PC2 (26.9%)
Site
SRS5 (healthy)
SRS6 (healthy)
BL60 (regen.)
CP40 (ghost)
Depth (cm)
90
45
15
0

(e)


CH4 (μM)
Salinity (PSU)
r = −0.44, p = 0.21
r = −0.03, p = 0.93
r = 0.62, p = 0.02
Campaign
Oct 2025
dry (Mar 2023)
wet (Oct 2022)
Disturbance
ghost
healthy
regenerating